

FORTIGATE NSE4

Lab 6 — Destination NAT

Virtual IPs and Port Forwarding to a Published Web Server

1. Source NAT and Destination NAT

Both translate addresses, but they solve opposite problems and sit on opposite sides of a session.

	Source NAT (SNAT)	Destination NAT (DNAT)
What it rewrites	The source address of outbound traffic	The destination address of inbound traffic
Direction	Inside → outside	Outside → inside
Problem solved	Private addresses are not routable externally	Internal servers have no externally reachable address
FortiOS object	The NAT toggle on a firewall policy	A Virtual IP object
Policy field	Configured within the policy itself	Referenced as the policy destination

This lab implements DNAT — commonly called a Virtual IP or port forwarding. An internal web server with a private address is published to a different network, so a client can reach it by connecting to an address on the firewall.

Two objects are required: a VIP defining the translation, and a firewall policy permitting the traffic. Neither works alone, and the failure mode when one is missing looks identical to a misconfiguration of the other.

2. Lab Topology

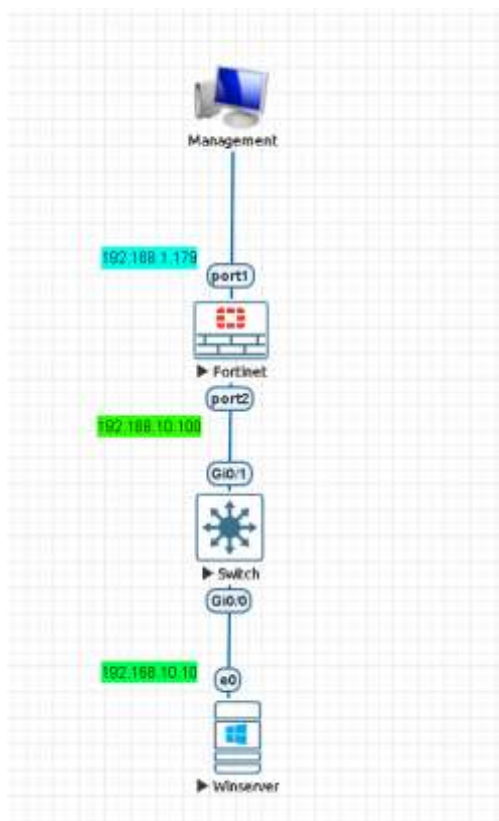


Figure 1 — Lab topology: the client reaches the server only through the FortiGate

Element	Address	Role
port1	192.168.1.179/24	External side. Where the client connects
port2	192.168.10.100/24	Server zone. Also the server's default gateway
Management PC	192.168.1.6	The client
Windows Server	192.168.10.10/24	IIS web server
VIP external IP	192.168.1.179	The interface address itself — see 2.1

► 2.1 Using the Interface Address as the VIP External IP

The VIP's external address is port1's own IP rather than a spare address on the subnet. This is legitimate and common — it is how a site with a single public IP publishes an internal server.

3. Preparing the Management Path

The FortiGate's administrative HTTP service listens on port 80 by default. Since the VIP will claim port 80 on the same address, the admin port is moved to 8080 before anything else is configured.

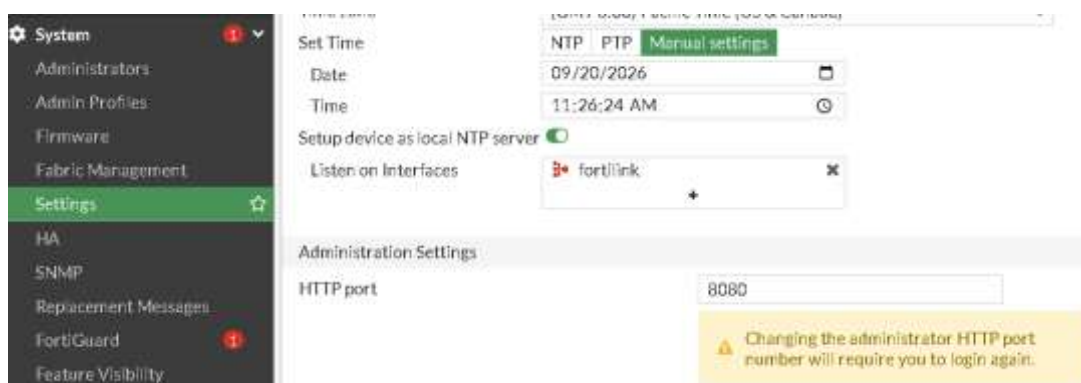


Figure 2 — Administrative HTTP port changed from 80 to 8080

```
config system global
    set admin-port 8080
end
```

This is a global setting, not per-interface, and the GUI warns that the change forces a re-login. Management moves to `http://192.168.1.179:8080`; HTTPS administration on 443 is unaffected.

► 3.1 Why This Ordering Matters

Configuring the VIP first would create a silent conflict on port 80 between the appliance's own admin service and the published server. Worse, a mistake in the VIP's external address would send the browser to the FortiGate login page instead of the web server — a confusing symptom that looks like a translation failure but is not.

Separating the management port from the published service port removes the ambiguity entirely.

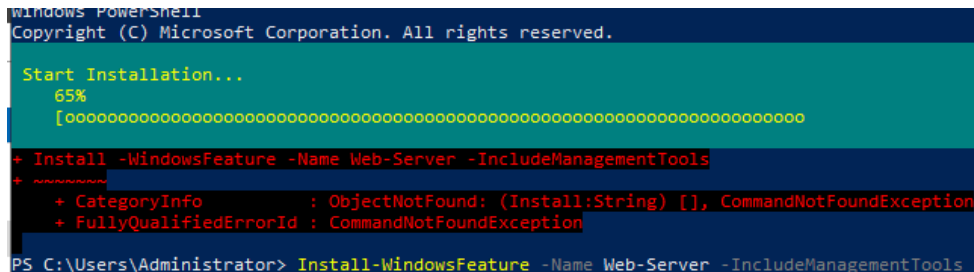
Interface	Physical Interface	IP Address	Role
LAN-Network (port2)	Physical Interface	192.168.10.100/255.255.255.0	PNP
port1	Physical Interface	192.168.1.179/255.255.255.0	HTTP
port3	Physical Interface	0.0.0.0/0.0.0.0	PNP
port4	Physical Interface	0.0.0.0/0.0.0.0	

Figure 3 — Interface list after addressing: port1 external, port2 as the server zone

4. Preparing the Web Server

IIS is installed on the Windows Server from PowerShell.

```
Install-WindowsFeature -Name Web-Server -IncludeManagementTools
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Start Installation...
65%
[oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo]

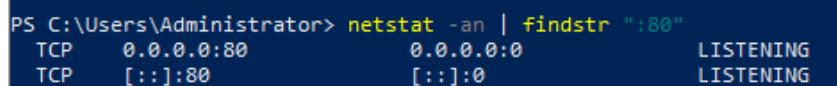
+ Install -WindowsFeature -Name Web-Server -IncludeManagementTools
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Install:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Administrator> Install-WindowsFeature -Name Web-Server -IncludeManagementTools
```

Figure 4 — Installing the Web-Server role

► 4.1 Confirming the Listener

```
netstat -an | findstr ":80"
```



```
PS C:\Users\Administrator> netstat -an | findstr ":80"
TCP 0.0.0.0:80 0.0.0.0:0 LISTENING
TCP [::]:80 [::]:0 LISTENING
```

Figure 5 — IIS listening on all interfaces

The result shows `0.0.0.0:80` in LISTENING state — IIS is bound to every interface. This is the value that matters.

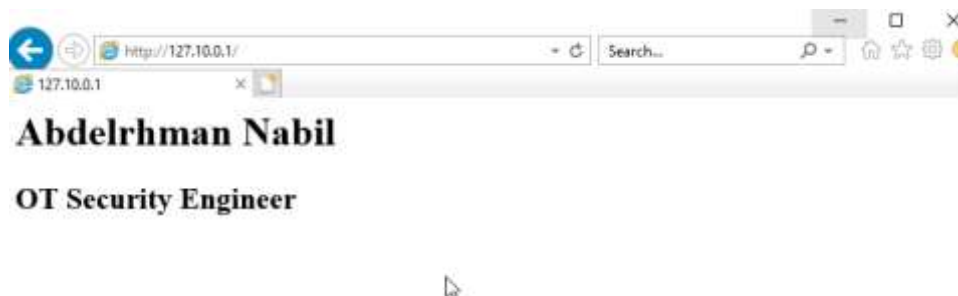


Figure 6 — The page served correctly from the server's own loopback address

This confirms IIS is running and serving content. It does not confirm network reachability — loopback traffic never touches the NIC or Windows Firewall — so it is a starting point rather than a result.

5. The Virtual IP Object

The screenshot shows the 'New Virtual IP' configuration window in FortiGate. The 'VIP type' is set to 'IPv4'. The 'Name' is 'to-Webpage'. The 'Comments' field is empty. The 'Color' is set to green. Under the 'Network' section, the 'Interface' is 'port1', the 'Type' is 'Static NAT', the 'External IP address/range' is '192.168.1.179', and the 'Map to' 'IPv4 address/range' is '192.168.10.10'. The 'Optional Filters' section has 'Port Forwarding' enabled. Under 'Port Forwarding', the 'Protocol' is 'TCP', the 'Port Mapping Type' is 'One to one', the 'External service port' is '80', and the 'Map to IPv4 port' is '80'.

Figure 7 — The VIP: external 192.168.1.179:80 mapped to 192.168.10.10:80

Field	Value	Meaning
Interface	port1	Where the FortiGate listens for this translation
Type	Static NAT	One-to-one address translation
External IP	192.168.1.179	What the client connects to
Map to	192.168.10.10	The real server
Port Forwarding	Enabled	Restricts the VIP to specific ports rather than all of them
Protocol	TCP	—
External service port	80	The port the client connects to
Map to IPv4 port	80	The port the server listens on

Enabling it scopes the VIP to TCP/80 alone. Ports 8080, 443, and 22 remain with the FortiGate regardless of what policies are written later — which is what prevents a future policy using this VIP with service ALL from hijacking the management session.

6. Baseline: the VIP Alone Is Not Enough

With the VIP created but no policy referencing it, the page was requested from the management PC.

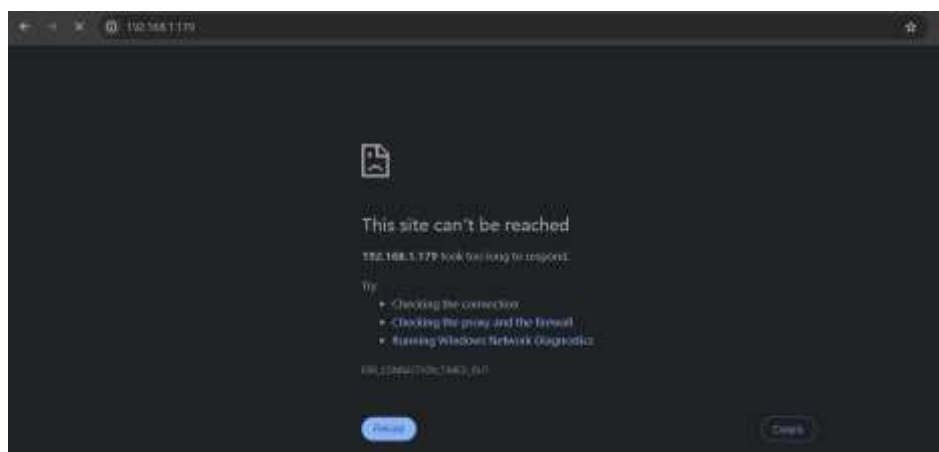


Figure 8 — The connection times out — the VIP exists, but no policy permits the traffic

7. The Firewall Policy

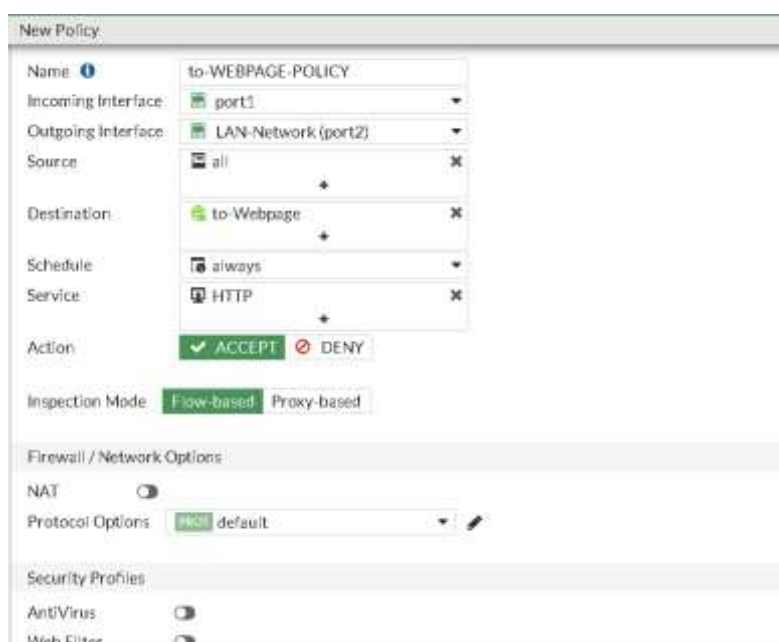


Figure 9 — The policy permitting HTTP from port1 to the server zone, with the VIP as destination

Field	Value	Reasoning
Incoming / Outgoing	port1 → LAN-Network (port2)	External side to server zone
Source	all	Any client on the external side
Destination	to-Webpage (the VIP)	The VIP object, not the server's address — see 7.1
Service	HTTP	Scoped to the published protocol
Action	ACCEPT	—
NAT	Disabled	Deliberate — see 7.2

► 7.1 The Destination Field Is Where This Usually Fails

The destination must be the VIP object. Setting it to an address object holding the server's real IP — or to the outgoing interface's subnet — causes the policy never to match, and the connection times out exactly as it did in the baseline.

The reason is the order of evaluation. The VIP already contains the mapping, so the policy references the **external identity** the client used, not the internal address the traffic ends up at. Fortinet names this as a common mistake in its own documentation.

► 7.2 Why NAT Is Disabled

The NAT toggle on a firewall policy performs *source* NAT, which is a separate operation from the destination NAT the VIP is already performing. Enabling it would rewrite the client's source address to 192.168.10.100 — the FortiGate's own port2 address.

Consequence: the server would see every visitor as the firewall. IIS logs, any per-client rate limiting, and any source-based access control on the server would all be operating on a single address. Attribution is lost for no benefit, since the server's replies route back through the FortiGate regardless.

8. Result



Figure 10 — The published page, reached from the client at the external address

The same request that timed out before the policy now returns the page. The client connected to 192.168.1.179 and was served content from 192.168.10.10 without any knowledge of that address.

9. Port Translation

The external and internal ports are independent. To demonstrate this, the VIP's external service port was changed to an arbitrary value while the mapped port remained at 80.



Figure 11 — External service port 12345 mapped to internal port 80



Figure 12 — The page served at the translated port

► 9.1 An Important Behaviour This Test Revealed

The firewall policy's service remained HTTP — that is, TCP/80 — throughout, and the connection to port 12345 still matched and succeeded.

Policy service matches the mapped port, not the external port. The intuitive expectation is that a policy filters on the port the client actually used, since policy evaluation would logically precede translation. The observed behaviour is the reverse, and Fortinet documents it explicitly: incoming traffic matching a VIP is treated as destination NAT, and the routing and policy lookup are performed **after** that translation has been applied.

The practical rule is that where a port-forwarding VIP changes the destination port, the **mapped** port is what must be permitted in the policy service — **not the external service port**.

This cuts both ways. A policy scoped to the external port would fail to match and the service would appear unreachable; and a policy scoped to the mapped port permits any external port the VIP is configured to forward, so the VIP object rather than the policy is what constrains external exposure.

10. Verification

A browser rendering a page proves that something answered. It does not show what the firewall did. A packet capture on the FortiGate does.

```
diagnose sniffer packet any "host 192.168.10.10 and port 80" 4
```

```
FortiGate-VN04-KVM # diagnose sniffer packet any "host 192.168.10.10 and port 80" 4
Using Original Sniffing Mode
interfaces=[any]
filters=[host 192.168.10.10 and port 80]
0.984706 port2 out 192.168.1.6.50959 -> 192.168.10.10.80: psh 3984841818 ack 2317362573
0.984750 port2 out 192.168.1.6.50959 -> 192.168.10.10.80: psh 3984841818 ack 2317362573
0.907888 port2 out 192.168.1.6.64325 -> 192.168.10.10.80: syn 1927822968
0.907931 port2 out 192.168.1.6.64325 -> 192.168.10.10.80: syn 1927822968
0.908238 port2 in 192.168.10.10.80 -> 192.168.1.6.50959: ack 3984842759
0.912537 port2 in 192.168.10.10.80 -> 192.168.1.6.64325: syn 1985223831 ack 1927822969
0.912937 port2 out 192.168.1.6.64325 -> 192.168.10.10.80: ack 1985223832
```

Figure 13 — Packet capture showing the translated session on port2

► 10.1 Reading the Capture

Observation	What it confirms
Source is 192.168.1.6	The real client address, not 192.168.10.100. Source NAT is correctly disabled and the server will log the true origin
Destination is 192.168.10.10:80	The client connected to 192.168.1.179, so the FortiGate rewrote the destination. This is the DNAT
Traffic is on port2	The translated packet is leaving toward the server zone, confirming the routing decision
syn / syn-ack / ack	A complete three-way handshake — the server accepted the connection rather than merely receiving the packets
psh	HTTP payload in flight on an established session

Together these show the full path: the client's request arriving for the external address, the destination being rewritten, the packet forwarded out of the server-zone interface, and the server completing the handshake.

11. Summary

Objective	Result
Publish an internal server to an external network	Achieved — reachable at 192.168.1.179 with no client knowledge of 192.168.10.10
Confirm a VIP alone grants no access	Confirmed — the request timed out until a policy referenced the VIP
Preserve client source addresses	Achieved — capture shows the real client IP reaching the server
Demonstrate port translation	Achieved — external 12345 mapped to internal 80 with no server-side change
Verify the translation independently	Achieved — packet capture shows the rewritten destination and a completed handshake